2406ET006

Roll No. 2 3 U g b C e e 0 9 0 e B

Course: B. Tech Computer Engineering/
B. Tech. Computer Science and Engineering (AI/ML)
Subject: Applied Chemistry
Subject Code: ETCH-108
Semester: II

Time: 03 Hours **Max Marks: 70**

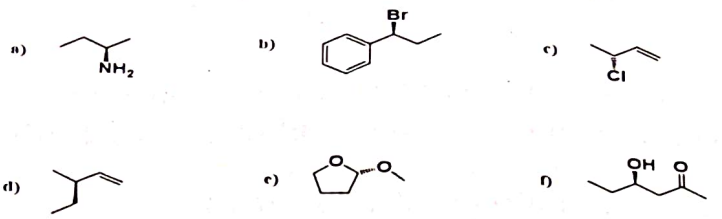
Instructions to the Students:

1. This Question paper consists of two Sections. All sections are compulsory.
2. Section A comprises 10 questions of short answer type. All questions are compulsory. Each question carries 2 marks.
3. Section B comprises 8 long answer type questions out of which students must attempt any 5. Each question carries 10 marks.
4. Do not write anything on the question paper.

Q. No.	SECTION –A (SHORT ANSWER TYPE QUESTIONS)	Marks
1. a.	Write down the electronic configuration of Cr^{3+} & Cu^{2+} [Cr (Z = 24); Cu (Z = 29)]	(2)
b.	He_2 molecule does not exist. Illustrate by using molecular orbital theory.	(2)
c.	Draw the functional isomers of an organic compound having chemical formula C_2H_6O and Metamers of an organic compound having chemical formula $C_4H_{10}O$.	(2)
d.	How to convert 2D Structure to 3D model in Chem-sketch?	(2)
e.	Calculate the crystal field stabilization energy (CFSE) for low spin and High Spin Fe2+. Do not include pairing energy	(2)
f.	Outline the use of Sci-finder software.	(2)
g.	Identify the type of reaction involved in Diels Alder Reaction. Show with one example.	(2)
h.	Categorize the type of reaction that happen in ozone layer by showing example.	(2)
i.	On the basis of IR Spectroscopy, Distinguish between Intermolecular and intramolecular Hydrogen Bonding.	(2)
j.	What happens to a molecule in an ionization chamber/section of a mass spectrometer?	(2)

SECTION –B (LONG ANSWER TYPE QUESTIONS)

2.	Draw the molecular orbital diagrams of Oxygen molecule (O2) and Nitrogen molecule (N2). Explain their magnetic nature and bond order.	(10)
3. a)	Differentiate between Configurations and Confirmations.	(4)
b)	Identify all the chiral centres in each molecule and determine the absolute configuration as R or S:	(6)

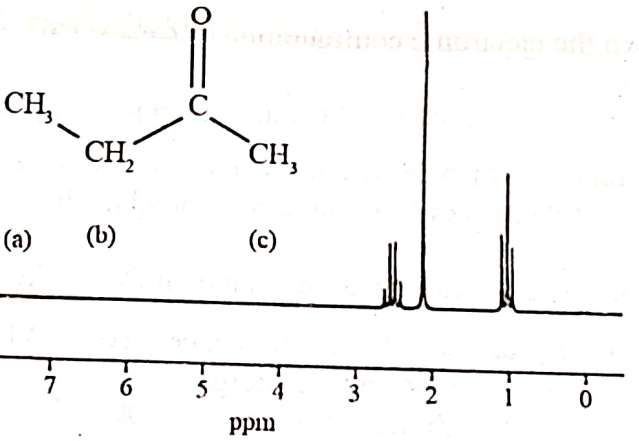


4. a) Derive Schrodinger wave equation? Explain the significance of the Ψ and Ψ^2. (5)
   b) Explain the application of Ψ and Ψ^2 to hydrogen atom. (5)

5. Draw the octahedral crystal field splitting diagrams for Fe2+ with both weak and strong octahedral fields. Label the diagrams weak and strong field, high spin and low spin, give the names of the d-orbitals, and label the appropriate orbital sets e_g and t_{2g} (10)

6. a) Draw the band diagrams of Conductors, Semiconductors and Insulators. (5)
   b) What is doping? Explain the role of doping on band structures. (5)

7. a) Read and assign the peaks of the 1H NMR spectrum of Butanone, shown below, and explain the spectrum in as much detail as you can, including the relative positions of the signals, the splitting and the relative sizes of the peaks (5)



   b) Sketch pi- molecular orbital of Benzene explaining its aromatic behaviour through Huckel's Rule.

8. a) Enlist applications of AI-ML in Material quality Control. (5)
   b) Explain the synthesis of Paracetamol by conventional and green route from phenol. (5)

9. a) What is the principle of IR spectroscopy? Calculate the fundamental modes of vibrations in C_6H_6 and CH_4 molecules. (5)
   b) Differentiate between fluorescence and phosphorescence. (2)
   c) Demonstrate prediction of IR Spectrum by using computers. (3)

===END OF PAPER===